Name: Meggie Chen

Date turned in: 3/3/26

Date due: 3/3/26

PSET 4

Acknowledgement: N/A

1. **Given:** $k_g = 1.5\ W/m\cdot K$ $\quad h_{rad} = 5\ W/m^2K$

$k_a = 1\ W/m\cdot K$

$\eta = 0.553 - 0.001\,T_{Si}$

(contact) $R_c = 10^{-4}\ m^2\cdot K/W$

No R_c between adhesive & silicon

$G = 800\ W/m^2K$

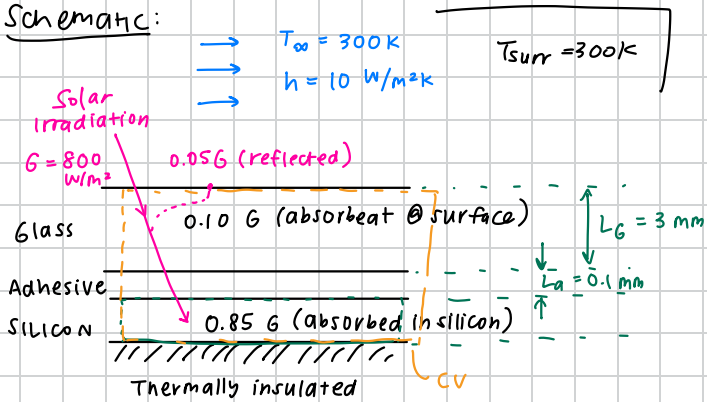
5% G reflected } top of glass

10% G absorbed

85% G absorbed @ silicon

electricity ← ↳ thermal E

Schematic:



Assume: steady state

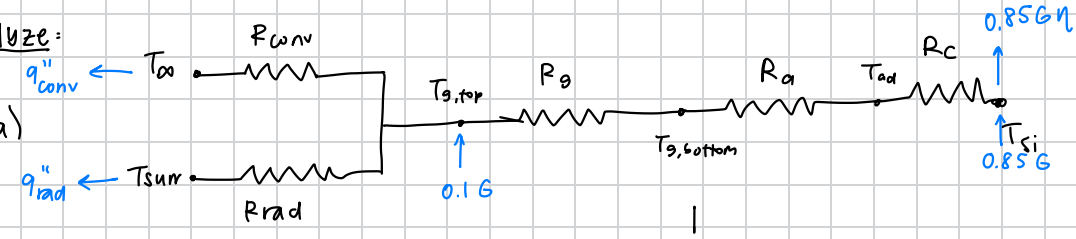
Find: (a) show thermal resistance network (R, nodes, heat fluxes, assumptions)

(b) $T_{g,top}$, T_{Si}

(c) η & Electricity produced

Analyze:

(a)



Assumptions:

① steady state

② 1-D heat transfer

③ constant mat props

④ Silicon bottom perf. insulated

⑤ No ga contact resistance

why is radiation only @ g surface?

$$R''_{conv} = \frac{1}{h} = \boxed{0.1\ m^2K/W}$$

$$R''_{rad} = \frac{1}{h_{rad}} = \frac{1}{5} = \boxed{0.2\ m^2K/W}$$

$$R''_g = \frac{L_g}{k_g} = \frac{0.003}{1.5} = \boxed{0.002\ m^2K/W}$$

$$R''_a = \frac{L_a}{k_a} = \frac{0.0001}{1} = \boxed{0.0001\ m^2K/W}$$

$$R''_c = \boxed{0.0001\ m^2K/W}$$

(b) CV of silicon:

$$q''_{in} = 0.85\,G(1-\eta) \Rightarrow \text{thermal energy}$$

$$q''_{out} = \frac{T_{Si} - T_{g,top}}{R_c + R_a + R_g} = \frac{T_{Si} - T_{g,top}}{0.0022}$$

$$680(1 - 0.553 + 0.001\,T_{Si}) = \frac{T_{Si} - T_{g,top}}{0.0022}$$

CV of s+a+g:

$$q''_{in} = 0.85G(1-\eta) + 0.1G$$

$$q''_{out} = \frac{T_{g,top} - T_\infty}{R_{surf}}$$

$$R_{surf} = \frac{1}{\frac{1}{0.1} + \frac{1}{0.2}}$$

$$680(1 - 0.553 + 0.001\,T_{Si}) + 80 = \frac{T_{g,top} - 300}{1/15}$$

$$\boxed{\begin{aligned} T_{g,top} &\approx 341.56\ K \\ T_{Si} &\approx 342.7374\ K \end{aligned}}$$

(c) $0.553 - 0.001(342.7374) = \eta = \boxed{0.21}$

$P'' = \eta(0.859) = 0.21(680)$

$\boxed{P'' \approx 142.98 \ W/m^2K}$

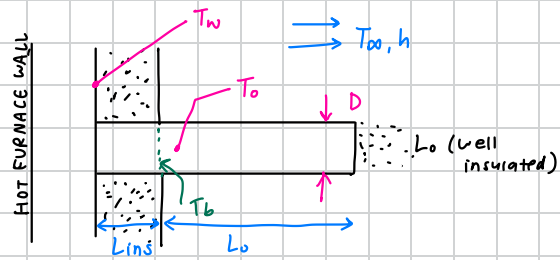
2. **Given:** $D = 10\ mm$

$k = 80\ W/m\cdot K$

$T_w = 200°C$, $T_o < T_{max} = 100°C$

$L_{ins} = 200\ mm$

$T_\infty = 20°C$, $h = 20\ W/m^2K$



Find: (a) $T_o = f(\text{temps})$ → eq.

(b) if $L_o = 200mm$, T_o ?

$T_o < T_{max}$?

(c) $(L_o)_{min}$ where $T_o < T_{max}$

Assumptions: steady state

constant mat. props

1-D

Analyze:

(a) $q = \frac{kA_c}{L_{ins}}(T_w - T_b)$ (insulation)

$q_f = M \tanh mL$

$= \sqrt{hPKA_c}\,(T_b - T_\infty)\tanh mL$ } fin

$\Rightarrow$ $\frac{kA_c}{L_{ins}}(T_w - T_b) = \sqrt{hPKA_c}\,(T_b - T_\infty)\tanh mL$

$$\boxed{T_b = \frac{\frac{kA_c T_w}{L_{ins}} + \sqrt{hPKA_c}\,T_\infty \tanh mL}{\frac{kA_c}{L_{ins}} + \sqrt{hPKA_c}\tanh mL}}$$

↳ $m = \sqrt{\frac{hP}{kA_c}}$

(b) $A_c = \pi(0.005)^2 = 7.854\times10^{-5}\ m^2$

$P = \pi(0.01) = 0.0314\ m$

$m = \sqrt{\frac{20(0.0314)}{80(7.854\times10^{-5})}} \cong 10$

$\frac{kA_c}{L_{ins}} = \frac{80(7.854\times10^{-5})}{0.2} \cong 0.031416$

$\tanh(mL_o) = \tanh(2) \approx 0.964$

$M = \sqrt{20(0.0314)(80)(7.854\times10^{-5})} \cong 0.062816$

$$T_b = \frac{0.031416(473) + 0.062816(0.964)(293)}{0.031416 + 0.062816(0.964)} \approx \boxed{354.5K}$$

→ 81.5°C

This is under T_{max}.

(c) Assume $T_b = 100°C = 373\ K$

$$373 = \frac{0.031416(473) + 0.062816(293)\tanh(10L)}{0.03146 + 0.062816\tanh(10L)}$$

$\tanh(10L) = \frac{3125}{5.025} = 0.622$

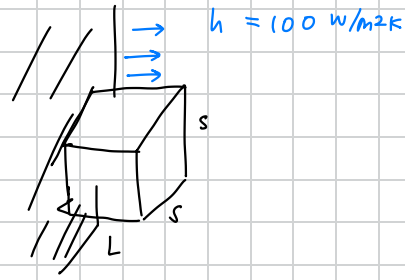
$10L \approx 0.733 \rightarrow L \approx 0.0733\ m$

$= \boxed{73.3\ mm}$

3. **Given:** $k = 400$ W/m·K
$s = 10$ mm
$h = 100$ W/m²K
$\eta_f = 0.6$

convection tip

Schematic:



Assume: steady state
constant material props

Find: (a) L
(b) R_f & ε_f

(a) $$\eta_f = \frac{\tanh mL_c}{mL_c}$$

$$m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{(100)(4\cdot 0.01)}{(400)(0.01)^2}} \approx 10$$

$$L_c = L + \frac{t}{2} = L + \frac{0.01}{2} = L + 0.005$$

$$0.6 = \frac{\tanh(10(L+0.005))}{10(L+0.005)}$$

$$L \approx 0.1462 \text{ m} = \boxed{146.2 \text{ mm}}$$

$$0.1405 \text{ m}$$

(b) $$R_f = \frac{1}{0.6(100)(A_f)}$$
$A_f \to 2wL_c$

$$= \frac{1}{60(2(0.01)(0.1462+0.005))} \cong \boxed{5.51 \text{ K/W}}$$

$$\varepsilon_f = \frac{q_f}{q_{no\ fin}} = \frac{\eta_f \cancel{k} A_f \cancel{\theta_b}}{\cancel{k} A_c \cancel{\theta_b}} = 0.6\left(\frac{2\cancel{w}L_c}{w^{\cancel{2}}}\right)$$

$$= 1.2\left(\frac{0.1462+0.005}{0.01}\right)$$

$$= \boxed{18.144}$$